

Driving A Wireless Robotic Car USING IR SENSORS

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Presented here is an Arduino and AT89C2051 microcontroller (MCU) based robot. The robot is controlled using infra-red (IR) sensors and wireless radio frequency (RF) communication. In this project we use four pairs of IR

sensor modules (IR LED and photodiode) to control forward, backward, left and right movements of the robot. The prototype of the robot is shown in Fig. 1 and IR modules mounted on four general-purpose PCBs in Fig. 2.

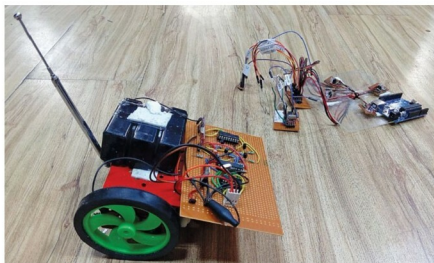


Fig. 1: Prototype of the robot

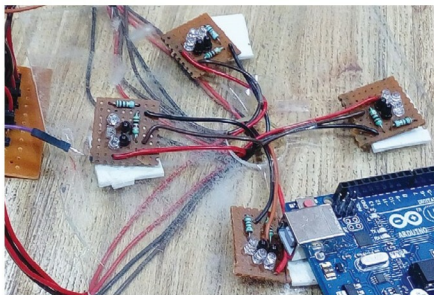


Fig. 2: Four pairs of IR modules

Circuit and working

Two separate circuits are used in this project; transmitter circuit as shown in Fig. 3 and receiver circuit

TABLE I
IR Pairs and Controls

IR Pair	Robot movement
IRRX1-IRTX1 (first pair)	Right
IRRX2-IRTX2 (second pair)	Left
IRRX3-IRTX3 (third pair)	Backward
IRRX4-IRTX4 (fourth pair)	Forward

PARTS LIST

Semiconductors:

- IC1, IC3 - 7805 voltage regulator
- IC2 - HT12E encoder
- IC4 - HT12D decoder
- IC5 - AT89C2051 MCU
- IC6 - L293D motor driver
- LED13, LED14 - 5mm LED
- IRTX1 - 5mm IR LED (LED1-LED3)
- IRTX2 - 5mm IR LED (LED4-LED6)
- IRTX3 - 5mm IR LED (LED7-LED9)
- IRTX4 - 5mm IR LED (LED10-LED12)
- IRRX1-IRRX4 - 5mm photodiode pair
- T1-T4 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 750-kilo-ohm
- R2-R9, R11, R13, R14-R23 - 330-ohm
- R10 - 33-kilo-ohm
- R12 - 10-kilo-ohm

Capacitors:

- C1 - 0.1 μ F ceramic
- C2, C3 - 22pF ceramic

Miscellaneous:

- BOARD1 - Arduino Uno board
- XTAL1 - 11.0592MHz crystal oscillator
- M1, M2 - DC-gear motor, 100rpm
- TX1 - 434MHz RF transmitter module
- RX1 - 434MHz RF receiver module
- CON1, CON3 - 2-pin connector terminal
- CON2 - 10-pin connector
- BATT1, BATT2 - 12V battery
- ANT.1, ANT.2 - 17cm-long single-strand wire antenna
- USB A-B cable